Your grade: 100%

Your latest: **100%** • Your highest: **100%** • To pass you need at least 80%. We keep your latest score.

Next item →

Instructions

1. Which of the following are key applications of deep learning in cybersecurity? Select all that apply. 1 / 1 point

- [x] Phishing email detection

  Nice work

  Correct! Deep learning can be used to analyze email content, detecting patterns associated with phishing attempts.

- [ ] Developing unbreakable encryption algorithms
- [x] Automated threat response

  Nice work

  Correct! Deep learning can automatically respond to detected threats, which helps reduce response time.

- [ ] Eliminating the need for firewalls
- [x] Advanced malware detection

  Nice work

  Correct! Deep learning models are capable of identifying and detecting advanced malware through pattern recognition and data analysis.

2. What are some advantages of using AI to support cybersecurity analysts? Select all that apply. 1 / 1 point

- [ ] Reduced need for trained cybersecurity professionals
- [ ] Complete elimination of cyber threats
- [x] Faster detection of security breaches

  Nice work

  Correct! AI speeds up the analysis process, allowing quicker identification of potential breaches.

- [x] Enhanced prediction of potential threats

  Nice work

  Correct! AI can analyze patterns to predict potential threats, providing a proactive defense.

- [x] Greater accuracy in threat detection

  Nice work

  Correct! AI can improve detection accuracy by reducing false positives.

3. A cybersecurity analyst implements an AI-based system for threat detection, noting its adaptability to new malware types. Which aspect of AI is most responsible for this adaptability? 1 / 1 point

- ( ) Dependence on human intervention for real-time decisions.
- (•) Machine learning algorithms that adapt and learn from new data.
- ( ) The ability to manually update rules within the system.
- ( ) Reliance on predefined threat signatures.

Nice work

Correct! Machine learning algorithms allow the system to identify and respond to novel malware types.

4. What factor has primarily driven the increased role of AI in cybersecurity in recent years? 1 / 1 point

- ( ) Reduced frequency of cyber threats.
- (•) Advancements in computational power and data availability.
- ( ) Decline in traditional cybersecurity costs.
- ( ) Government regulations mandating AI use in cybersecurity.

Nice work

Correct! Greater computational power and large datasets have significantly contributed to AI's expanded role in cybersecurity.

5. Which statement best describes the historical development of AI in cybersecurity? 1 / 1 point

- ( ) AI was first introduced to cybersecurity in the 2000s.
- (•) AI applications in cybersecurity began in the 1980s, with recent advancements driving renewed interest.
- ( ) AI applications in cybersecurity preceded other industry uses.
- ( ) AI gained popularity in cybersecurity in the 1990s due to a decrease in cyber threats.

Nice work

Correct! AI was initially explored for cybersecurity in the 1980s, but recent advances have led to a resurgence.